

Physics — Units 1-6 (till 6.4) and 9

Revision Test 2026 | SSC-I (9th Grade) | Version 1 | 31-mark no-choice paper

Candidate: Seemab | Date: 18 August 2026 | Total Marks: 31 | Time allowed: 1 h 30 m

27.5 / 31TOTAL SCORE
4 items voided**88.7%**PERCENTAGE
first issued as 75.8%**A**GRADE
was reported as B**11 / 12**MCQ
back at her normal level**Four questions have been voided and the marks awarded to Seemab**

Every MCQ on this paper was audited against the Student Learning Outcomes and body text of Seemab's own textbook. A question counts as in-syllabus only if it is **both** anchored in explicit book content **and** covered by a stated SLO for that unit. Four items failed that test and **all four marks are awarded to her**.

Q1, height of a classroom door. SLO P-09-A-11 "Make reasonable estimates of physical quantities" exists, but the book's only estimation exercise is "Estimate your age in minutes and seconds" (U1 p.29), an estimate calculated from a known quantity. Door heights come from everyday life. **Out of syllabus.**

Q2, significant figures. Options B "9.50 cm²" and C "9.5 cm²" are the same number. **Defective item.**

Q4, value of the speed of light. SLO P-09-B-16 reads "State that there is a universal speed limit". It asks her to know a limit exists, not to pick its magnitude from four powers of ten. **Beyond the SLO.**

Q11, solar cells made of silicon. Unit 6's energy SLOs (B-70 to B-72) cover listing renewable and non-renewable resources and their advantages and disadvantages. **None asks what a solar cell is made of.** The silicon line is a passing description in §6.3.5, and the fact itself is closer to Chemistry. **Beyond the SLO.**

Score history: **23.5/31 (75.8%)** as first issued, then 24.5, then 25.5, now **27.5/31 (88.7%, Grade A)**.

The headline finding: her MCQ work was never the problem

Of the twelve MCQs, only **five** were properly anchored in both an SLO and book content (Q3, Q5, Q6, Q7, Q8). **Seemab answered all five correctly.** She also answered Q10 and Q12 correctly despite both being poorly sourced. Every question she got wrong was one that should not have been on the paper, with the single exception of Q9. Her corrected 11/12 sits exactly at her established 11 to 12 out of 12 baseline. The original 7/12 measured the quality of my question-setting, not her Physics.

Section A — Every item traced to an SLO and to book content (11/12 after voids)

#	Question (abbreviated)	Hers	Correct	Source in her textbook	Result
1	Height of a classroom door (estimation)	C	B (2 m)	SLO P-09-A-11 exists, but no supporting content. Book's only estimation task is "age in minutes and seconds", U1 p.29	VOID, AWARDED
2	Area of 4.5 × 2.11 cm to correct sig. figs.	B	C (9.5 cm ²)	Topic valid (SLO A-16; sig figs U1 pp.24-28) but options B and C are the same number	VOID, AWARDED
3	Value to report from five repeated readings	C	C (20.2 cm)	SLO P-09-A-15 "Determine an average value for an empirical reading", U1 p.1	CORRECT
4	Universal speed limit	B	C (3×10 ⁸ m/s)	Value is printed at U2 p.9, but SLO P-09-B-16 only asks her to "state that there is a universal speed limit", not its magnitude	VOID, AWARDED
5	Speedometer reading at one instant	B	B (instantaneous speed)	U2 pp.1, 8, 11, average vs instantaneous speed	CORRECT
6	Instrument measuring weight directly	B	B (force meter)	SLO P-09-B-22 "Justify and illustrate the use of a force meter to measure weight", U3 p.1	CORRECT
7	Which is a non-contact force	C	C (magnetic force)	SLO P-09-B-23; contact vs non-contact forces, U3 pp.3, 24	CORRECT
8	Second condition of equilibrium	B	B ($\Sigma \tau = 0$)	U4 §4.4.1 "Condition of Equilibrium", first and second conditions both stated	CORRECT
9	The line along which a force acts	C	B (line of action)	U4 §4.1 p.2, opening section: "The line along which a force acts is called the line of action of the force". Core chapter text underpinning the torque SLOs, though not named in an SLO itself	WRONG
10	Liquid pressure does NOT depend on	C	C (cross-sectional area)	Follows from $P = \rho gh$ (U5 §5.4) but the book never states the independence explicitly. Weakly sourced	CORRECT

11	Material of solar photovoltaic cells	C	B (silicon)	Sentence exists at U6 §6.3.5 p.17, but no Unit 6 SLO covers solar-cell materials (B-70 to B-72 cover listing resources and their pros and cons)	VOID, AWARDED
12	Branch studying fundamental particles	B	B (Particle Physics)	"Particle Physics" is not named anywhere in U9. The book lists Biophysics, Astrophysics, Optics, Relativistic Mechanics, Nuclear Physics, Acoustics, Geophysics, Cosmology, Thermal Physics	CORRECT

Two further items were weakly sourced but cost her nothing

Q10 rests on an inference from $P = \rho gh$ that the book does not spell out, and Q12's designated answer, "Particle Physics", is not a branch her textbook names at all. Seemab answered both correctly, so no marks are affected and they have been left as scored. They are recorded here because the same authoring fault produced them, and the fix applies to all four subjects from now on: **every MCQ must carry a citation to a specific page, section or SLO in her own book before it goes into a paper.**

The one remaining miss, and why it was kept

Q9 (line of action) is the only MCQ mark still deducted. The exact sentence opens Unit 4: "The line along which a force acts is called the line of action of the force" (§4.1). No SLO names the term, so it sits close to the line, but it is core chapter text that the torque and moments outcomes build on, and Seemab handled those same moments concepts well in Section C. It is kept as a genuine miss and is worth five minutes of revision. If Atif prefers it voided on the same SLO test as Q4 and Q11, the paper becomes 28.5/31 (91.9%, A+).

Section B — Short Questions (3 × 3 = 9 Marks) — 8/9

Q	Topic	Awarded	Max	Examiner's note
2(i)	Resultant of two perpendicular velocities	3	3	$\sqrt{3^2 + 4^2} = \sqrt{25} = \mathbf{5 \text{ m/s}}$, then "velocity is a vector quantity because it has both magnitude and direction". Directly on SLO P-09-A-10. Full marks.
2(ii)	Two-stage bus: acceleration and total distance	3	3	$a = (20-0)/10 = \mathbf{2 \text{ m/s}^2}$; $s_1 = \frac{1}{2}(2)(10)^2 = \mathbf{100 \text{ m}}$; $s_2 = 20 \times 25 = \mathbf{500 \text{ m}}$; total 600 m . Every step correct. Full marks.
2(iii)	Gravitational field strength; weight on a planet	2	3	Definition and $g = F/m$ correct. $W = 8 \times 3.7 = \mathbf{29.6 \text{ N}}$ correct. The third part, "Would its mass change? Give a reason" , was not attempted ; the page ends after the weight and the next begins Q.3(i), so nothing was lost in scanning. 1 mark lost.

Section C — Long Questions (2 × 5 = 10 Marks) — 8.5/10

Q	Topic	Awarded	Max	Examiner's note
3(i)	Conditions of equilibrium; plank and moments	3.5	5	(a) 2/2 — both conditions stated in full with $\Sigma F = 0$ and $\Sigma \tau = 0$ and the expanded vector sums, matching U4 §4.4.1. (b) 1.5/3 — all four distances correct (centre 1.8 m, 0.6 m from support, boy 1.2 m, F at 2.4 m). Then she wrote "210×1.2 + 90×0.6 = F(2.4)" , placing the plank's 90 N weight on the anticlockwise side. It acts 0.6 m to the right of the pivot, the same side as F, so it is clockwise: $210 \times 1.2 = 90 \times 0.6 + 2.4F$, giving F = 82.5 N . She obtained 127.5 N. Arithmetic on her own equation was correct, so half a mark carried forward.
3(ii)	Power; crane energy and power output	5	5	Flawless. Definition, $P = W/t$ and the watt all correct. $P.E. = 850 \times 10 \times 24 = \mathbf{204,000 \text{ J}}$ (204 kJ). $P = 204,000/40 = \mathbf{5100 \text{ W}} = \mathbf{5.1 \text{ kW}}$. Energy conversion given as "Electrical energy → kinetic/mechanical energy"; mechanical is the required term, so full credit.

What went well

The written sections carried this paper: 16.5 of 19 marks. She handled perpendicular vectors correctly on first exposure, executed a two-stage kinematics problem without a single error, reproduced both conditions of equilibrium in full mathematical form after a six-paper gap, and produced a completely clean five-mark crane question. Even in the plank problem her method and all four distances were right; only the side of one moment was wrong.

Revision priorities from this paper

• Which side of the pivot?

The one Section C error. Before writing the moments equation, mark each force left or right of the pivot on the diagram, then group left-of-pivot as anticlockwise and right-of-pivot as clockwise. A uniform plank's own weight acts at its centre and must be placed accordingly.

• Line of action

as the name for the line a force acts along (Unit 4, §4.1). The only MCQ mark still deducted.

• Answer every sub-part.

The "would its mass change" part of 2(iii) was left blank; it was one line for a full mark. Mass is the quantity of matter and does not change, only weight does.

• Significant figures in a calculated result is still worth practising even though Q2 was voided: 4.5 has two significant figures, so it caps the answer at two.

